



Classification

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Lecture Outline

- Classification: Why not Linear Regression?
- Binary Response & Logistic Regression
- Estimating the Simple Logistic Model
- Classification using the Logistic Model
- Multiple Logistic Regression



Heart Data

response variable
is Yes/No

Age	Sex	ChestPain	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	Thal	AHD
63	1	typical	145	233	1	2	150	0	2.3	3	0.0	fixed	No
67	1	asymptomatic	160	286	0	2	108	1	1.5	2	3.0	normal	Yes
67	1	asymptomatic	120	229	0	2	129	1	2.6	2	2.0	reversible	Yes
37	1	nonanginal	130	250	0	0	187	0	3.5	3	0.0	normal	No
41	0	nontypical	130	204	0	2	172	0	1.4	1	0.0	normal	No



Heart Data

This data contains a binary outcome HD for 303 patients who presented with chest pain. An outcome value of:

- **Yes**, indicates the presence of heart disease based on an angiographic test
- **No**, means no heart disease.

There are 13 predictors including:

- Age
- Sex (0 for women, 1 for men)
- Chol (a cholesterol measurement),
- MaxHR
- RestBP

...and other heart and lung function measurements.



Classification

Up to this point, the methods we have seen have centered around modeling and the prediction of a **quantitative** response variable (ex, number of taxi pickups, number of bike rentals, etc). Linear **regression** (and Ridge, LASSO, etc) performs well under these situations

When the response variable is **categorical**, then the problem is no longer called a regression problem but is instead labeled as a **classification problem**.

The goal is to attempt to classify each observation into a category (aka, class or cluster) defined by Y , based on a set of predictor variables X .



Typical Classification Examples

The motivating examples for this lecture(s), is mostly based on medical data sets. Classification problems are common in this domain:

- Trying to determine where to set the *cut-off* for some diagnostic test (pregnancy tests, prostate or breast cancer screening tests, etc...)
- Trying to determine if cancer has gone into remission based on treatment and various other indicators
- Trying to classify patients into types or classes of disease based on various genomic markers



Simple Classification Example

Given a dataset:

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

where the y are categorical (sometimes referred to as *qualitative*), we would like to be able to predict which category y takes on given x .

A categorical variable y could be encoded to be quantitative. For example, if y represents concentration of Clemson undergrads, then y could take on the values:

$$y = \begin{cases} 1 & \text{if Computer Science (CS)} \\ 2 & \text{if Statistics} \\ 3 & \text{otherwise} \end{cases}.$$

Linear regression **does not work well**,
or is not appropriate at all, in this setting.



Simple Classification Example

A linear regression could be used to predict y from x . What would be wrong with such a model?

The model would imply a specific ordering of the outcome and would treat a one-unit change in y equivalent. The jump from $y = 1$ to $y = 2$ (CS to Statistics) should not be interpreted as the same as a jump from $y = 2$ to $y = 3$ (Statistics to everyone else).

Similarly, the response variable could be reordered such that $y = 1$ represents Statistics and $y = 2$ represents CS, and then the model estimates and predictions would be fundamentally different.

If the categorical response variable was *ordinal* (had a natural ordering, like class year, Freshman, Sophomore, etc.), then a linear regression model would make some sense but is still not ideal.

Simple Classification Example: Binary Response

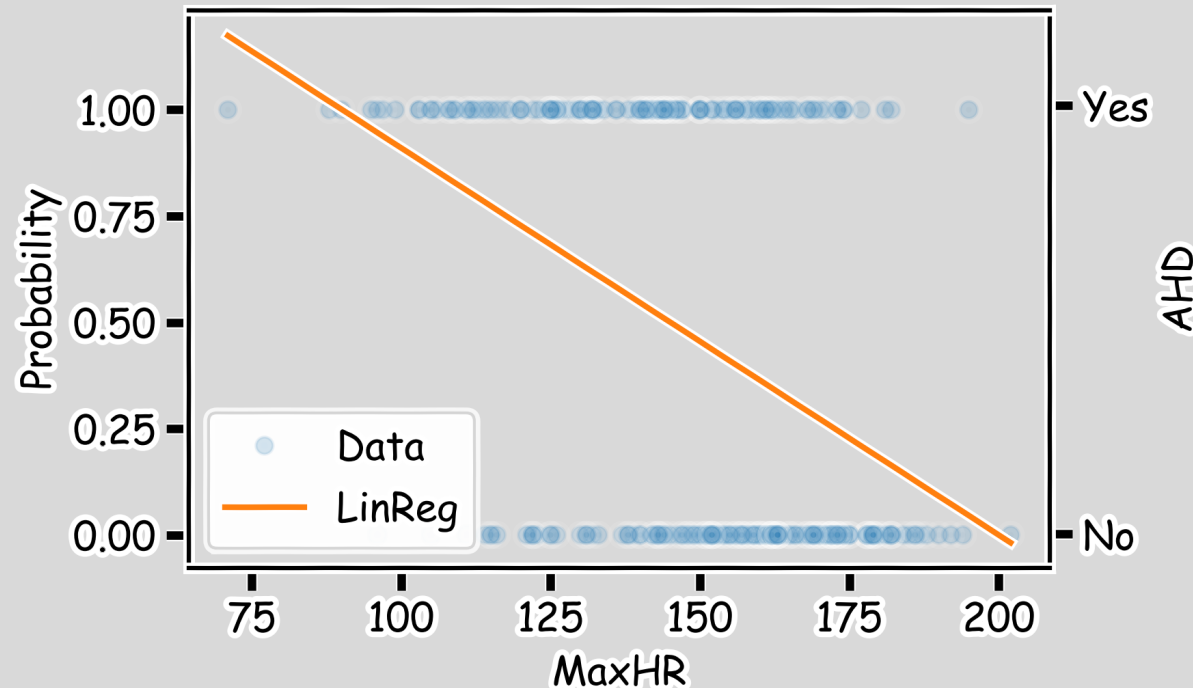
The simplest form of classification is when the response variable y has only two categories, and then an ordering of the categories is natural. For example, an upperclassmen Clemson student could be categorized as (note, the $y = 0$ category is a "catch-all" so it would involve both on-campus students and those who live in other situations: off campus, etc):

$$y = \begin{cases} 1 & \text{if } i \text{ lives in the Quad} \\ 0 & \text{otherwise} \end{cases}.$$

Linear regression could be used to predict y directly from a set of covariates (like sex, athlete or not, concentration, GPA, etc.), and if $\hat{y} \geq 0.5$, we could predict the student lives in the Quad and predict other houses if $\hat{y} < 0.5$.

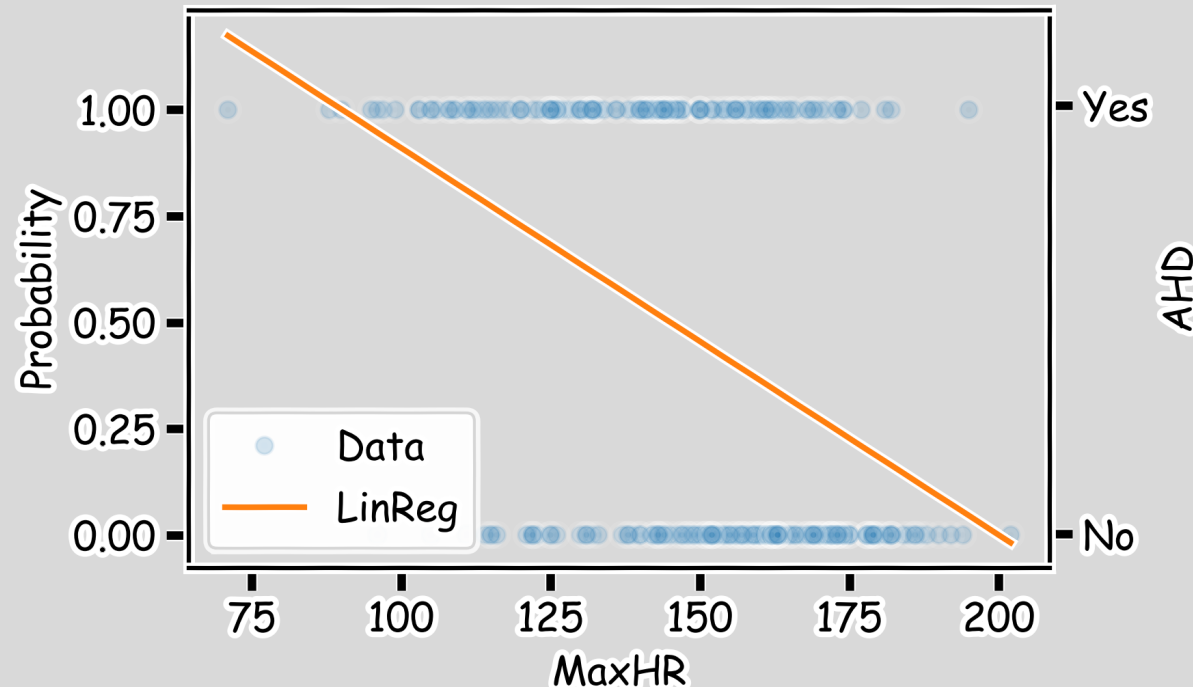
Simple Classification Example: Binary Response

What could go wrong with this linear regression model?



Simple Classification Example: Binary Response

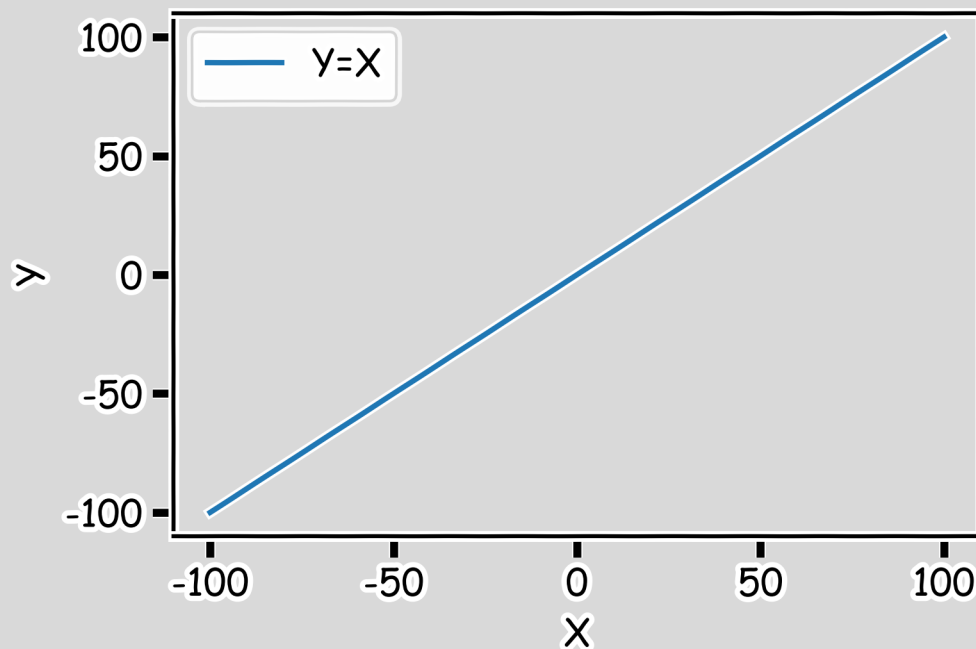
The main issue is you could get nonsensical values for y . Since this is modeling $P(y = 1)$, values for $\hat{y}$ below 0 and above 1 would be at odds with the natural measure for y . Linear regression can lead to this issue.





Simple Classification Example: Binary Response

Think of a function that would do this for us:



$$Y = f(x)$$

